

Course code: CSW 306
Course name: Backend Development
Type of Exam: Opened book / Practice on Computer

Exam code: 01
Duration: 120'
Maximum mark: 100

Instructions:

- Non-programming calculator: Yes ☒ | No ☐
- This paper consists of 5 questions. This paper carries 100 marks:
 - Question 1 (20 marks).
 - Question 2 (25 marks).
 - Question 3 (40 marks).
 - Question 4 (15 marks).

Students must create the project using the naming convention:

Format: `DateNow(DD-MM-YYYY)_BE_YourName`

Example: `26082025_BE_NguyenXuanCuong`

Students zip the project source code and include the database script or backup file in one compressed file for submission.

Students may refer to the .NET documentation to design and develop ASP.NET Core API web applications using the following:

1. <https://learn.microsoft.com/en-us/aspnet/core/getting-started/?view=aspnetcore-8.0>.
2. Print or pdf documents available.

HUMAN RESOURCE MANAGEMENT SYSTEM

Company XYZ wants to build an internal Web API system to manage employees and departments. You are required to develop a Web API to fulfill the following requirements.

Question 1: (20 marks) Database, Model and DbContext Design

- Create two classes: Employee and Department with a one-to-many relationship:
 - One department has many employees.
 - Each employee belongs to one department.
- Add Data Annotations for validation:
 - FullName: required, max 200 characters
 - Email: required, valid email format
 - StartDate: must not be in the future

- Create a class AppDbContext that inherits from DbContext, and declare DbSet<Employee> and DbSet<Department>.

Note: Create it in the database using data that corresponds to the class you have designed.

Question 2: (25 marks) Repository and Service Layer

- Create IEmployeeRepository and EmployeeRepository interfaces and classes with the following methods:
 - Task<List<Employee>> GetAll ();
 - Task<Employee?> GetById (int Id);
 - Task Add (Employee employee);
 - Task Update (Employee employee);
 - Task Delete(int Id);
- Create IEmployeeService and EmployeeService with similar methods.
- Implement logic to connect with AppDbContext in the repository.
- Add method to retrieve employees by department (GetByDepartmentId(int departmentId) or similar).

Question 3: (40 marks) API Controller

Create EmployeesController with the following endpoints:

Method	Route	Description
GET	/api/employees	Return list of all employees (with department name)
GET	/api/employees/{id}	Return employee by ID
POST	/api/employees	Create new employee (accept nested department object)
PUT	/api/employees/{id}	Update employee
DELETE	/api/employees/{id}	Delete employee

Requirements:

- Use dependency injection to inject IEmployeeService.
- Handle not found cases (404) for invalid **Route**.
- Validate input data.

Question 4: (15 marks) Advanced: Nested JSON Handling

- Allow nested JSON input when creating employees, for example:

```
{
  "fullName": "Nguyen Van A",
  "email": "a.nguyen@example.com",
```



```
"startDate": "2023-01-01",  
"department": {  
  "name": "IT"  
}  
}
```

- If the department "IT" does not exist, create a new one.
 - If it already exists, assign the employee to that department.
- Return employee data with nested department information when calling GET /api/employees.

-----The End -----

Lecturer's signature



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